SI 305 | Final Report

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1. Summary

This analysis looks at whether toxic carcinogen releases in 2005 have a correlation with Michigan cancer rates in the late 2010s on a per-county basis, using data from the EPA's Toxic Release Inventory and Michigan's publicly available cancer atlas. Although multivariate regression showed no direct relationship between carcinogen release and cancer rates, it suggested that smoking rate, poverty rate, and obesity rate are strongly correlated with cancer incidence. However, a one sample t-test showed that Midland county had a significantly higher cancer incidence rate when compared to demographically and socioeconomically similar counties, suggesting a potential public health concern.

2. Problem Diagnosis

Michigan has one of the largest concentrations of manufacturing and chemical processing facilities in the United States, many of which release substances known as probable human carcinogens (PHCs). According to the U.S. Environmental Protection Agency's Toxic Release Inventory, Michigan facilities reported releasing millions of pounds of onsite chemicals annually, with a significant amount of those chemicals flagged as carcinogens.[1] Despite this potential public health crisis, the relationship between cancer incidence and industrial carcinogen release on a per-county scale has not been widely examined in publicly available research. The Michigan Cancer Atlas lists reports of cancer incidence and mortality by county in regards to social, economic, and demographic factors, but does not link these patterns to industrial emissions.[5]

This public health problem is best illustrated by Midland County, the home of DOW Chemical Company's largest plant. Decades of dioxin manufacturing and irresponsible disposal contaminated the Tittabawassee River floodplain, one of the largest dioxin contaminated sites ever documented in the United States, affecting the health of residents who lived, fished, farmed, and spent time along the river.[2] Members of the Midland community and environmental advocacy groups have long speculated that cancer rates are elevated as a consequence, yet a rigorous, data-driven county comparison has not been widely undertaken in publicly accessible research.

Midland county is home to roughly 84,000 residents according to the 2020 US Census, with approximately 91 percent of the population being White and about 20 percent of the population aged 65 or older.[6] Those most directly affected by industrial carcinogens are blue collar workers and former workers of the DOW Chemical plant, as well as residents of neighboring communities and floodplain communities located the closest to the plant. Many members of these communities already experience barriers to healthcare access, compounding the harm.[3] On a larger scale, the 2005 TRI data shows carcinogen release across dozens of other Michigan counties, meaning the potentially affected population may even extend into the millions, well beyond the scope of Midland.

Cancer remains the second leading cause of death in Michigan, and the Michigan Department of Health and Human Services has documented meaningful variation in cancer incidence across counties, though the cause of that variation has not been fully explained.[4] [5] Without understanding whether and where industrial carcinogen exposure contributes to elevated cancer risk, policymakers, public-health officials, and affected communities lack the evidence needed to prioritize environmental remediation, regulatory enforcement, or targeted cancer-screening programs.

1. U.S. Environmental Protection Agency, *TRI National Analysis* (Washington, D.C.: EPA, updated annually), https://www.epa.gov/trinationalanalysis.
2. U.S. Environmental Protection Agency, Region 5, *Tittabawassee River, Saginaw River & Bay Superfund Site*, EPA Site ID: 0503250, https://cumulis.epa.gov/supercpad/SiteProfiles/index.cfm?fuseaction=second.cleanup&id=0503250; U.S. Fish & Wildlife Service, *Tittabawassee River Natural Resource Damage Assessment and Restoration* (Washington, D.C.: USFWS, 2020), https://www.fws.gov/project/tittabawassee-river-natural-resource-damage-assessment-and-restoration.
3. Paul Mohai and Robin Saha, "Which Came First, People or Pollution? A Review of Theory and Evidence from Longitudinal Environmental Justice Studies," *Environmental Research Letters* 10, no. 12 (2015): 125011, https://doi.org/10.1088/1748-9326/10/12/125011.
4. Jiaquan Xu, Sherry L. Murphy, Kenneth D. Kochanek, and Elizabeth Arias, *Deaths: Final Data for 2022*, National Vital Statistics Reports (Hyattsville, MD: National Center for Health Statistics, 2024), https://www.ncbi.nlm.nih.gov/books/NBK616549/.
5. Michigan Department of Health and Human Services, Michigan Cancer Surveillance Program, *Cancer Incidence and Mortality Tables* (Lansing, MI: MDHHS, updated through 2020), https://vitalstats.michigan.gov/osr/cancer/summary.asp.
6. U.S. Census Bureau, 2020 Decennial Census, Midland County, Michigan, https://www.census.gov/quickfacts/midlandcountymichigan.

3. Research Questions

1. RQ1: Do Michigan counties with higher traces of total carcinogen chemicals released in 2005 (as reported by the EPA Toxic Release Inventory) have higher overall cancer incidence rates from 2015-2019 (as reported by the Michigan Cancer Atlas), after accounting for key county level differences in smoking rate, obesity rate, poverty rate, and racial makeup?

2. RQ2: How does Midland County's cancer incidence rate compare to other Michigan counties with similar poverty rates and racial makeups, and is the observed difference significant?

4. Analysis for RQ1

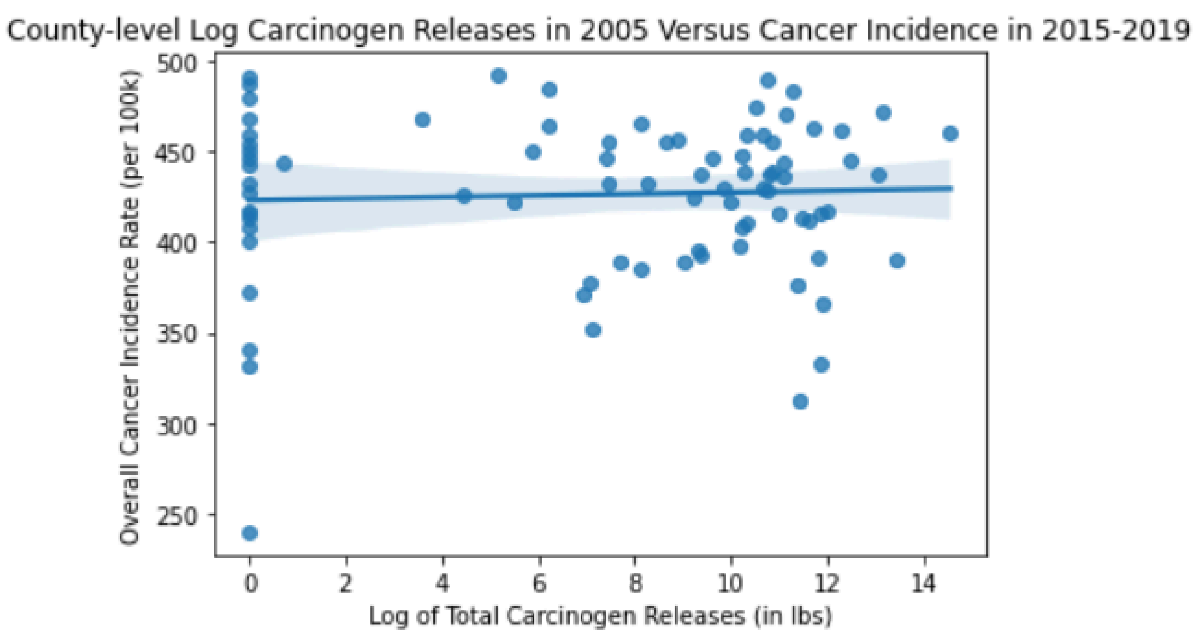
How I Analyzed the Data

To investigate the relationship between carcinogen releases and cancer incidence at a county level across Michigan, I first filtered the 2005 EPA Toxic Release Inventory dataset to only include records marked with carcinogen release, aggregating that with total releases by county. I then joined this to the Michigan Cancer Atlas, which provided cancer incidence rates, along with breakdowns of key demographic, social, and economic details by county. Since there were a small number of counties with a disproportionately high amount of carcinogen release, I applied a log transformation to reduce the impact of outliers. Then, I computed the Pearson correlation between log total carcinogen release and cancer incidence, followed by fitting a multivariate linear regression that added four control variables (poverty rate, percent non-white, obesity rate, and smoking rate).

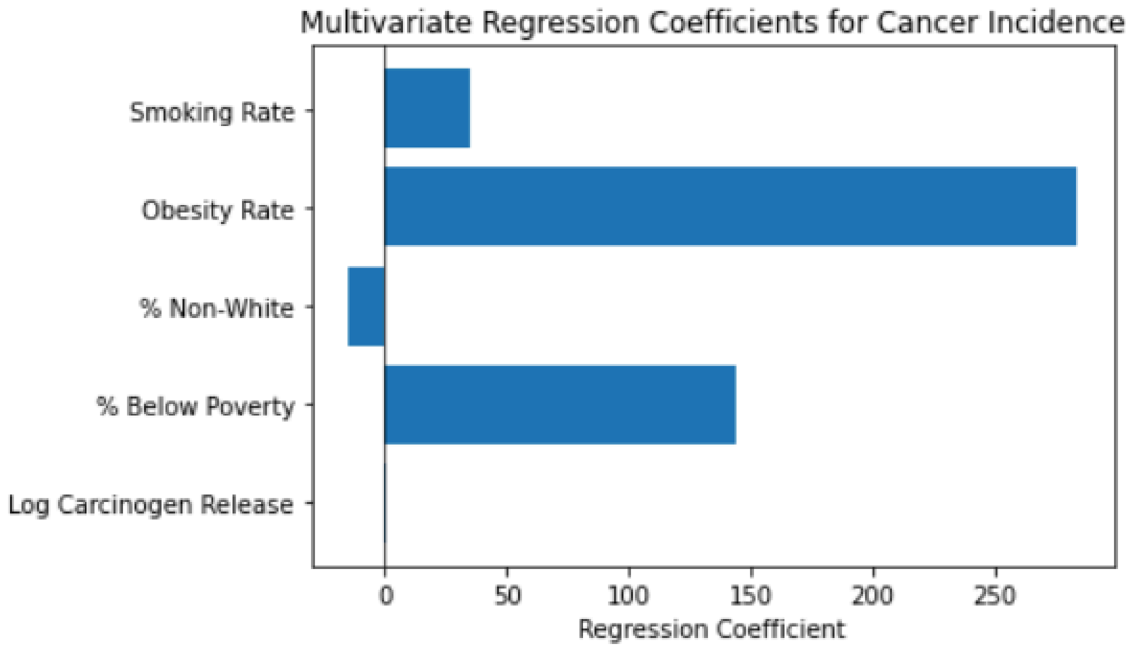
What I found

The analysis ultimately indicated no detectable or meaningful relationship between log total carcinogen release and cancer incidence at a county level. The Pearson r value was near zero ($r = 0.045$), and the p value was too large to suggest statistical significance ($p = 0.68$). These results also appeared in the multivariate regression. After adding controls, the algorithm's R^2 value remained extremely low, at 0.085. Along with this, the regression coefficient for log total release was approximately 0.046, only marginally higher than the previous analysis. By contrast, obesity rate had the largest coefficient at 283.7, followed by poverty rate at 144.2, then smoking rate at 35.4. These results were also mirrored when tested with log on-site carcinogen release rather than total release, and removing the Wayne County outlier, both of which produced very similar results. An important limitation of this analysis comes from the fact that it is derived from county data rather than individual data, so the results cannot be used to determine whether proximity to a pollution site increases cancer risk. Coupled with this, obesity and smoking rates were only available at a prosperity-region level rather than a county-level, limiting accuracy across the 83 counties. With only 83 observations and five predictor variables, degrees of freedom are tight, meaning the regression coefficients may not be causal. It should also be noted that counties that had no report of carcinogen release were converted from a null value to zero, meaning that some counties may not be completely void of carcinogens.

Visualizations



County-level log carcinogen releases in 2005 show no meaningful linear association with Cancer Incidence in 2015-2019 (Pearson r = 0.045, p = 0.68)



Obesity rate and poverty rate appear to be strong predictors of cancer incidence at a county-level, but log carcinogen release has a near zero coefficient, pointing to limited association.

Interpretation

These results suggest that at a county level, toxic carcinogen release alone does not meaningfully contribute to increased cancer incidence. Rather, social and socioeconomic factors such as obesity, poverty, and smoking appear to be a far more accurate predictor. That being said, these results are not entirely surprising. County level carcinogen release totals fail to account for actual human exposure, with much of their effects impacting wildlife and the environment (which can ultimately impact humans). Along with this, carcinogen release is not evenly spread across counties, meaning some individuals are impacted more than others, which is not reflected in the data. The near zero correlation coefficient does not mean carcinogen pollution is harmless, but rather that it has little effect on this scope of analysis. These findings motivated RQ2, which focuses on Midland county and asks whether cancer incidence is elevated when compared to similar counties, rather than looking for a pattern across all 83 counties.

5. Analysis for RQ2

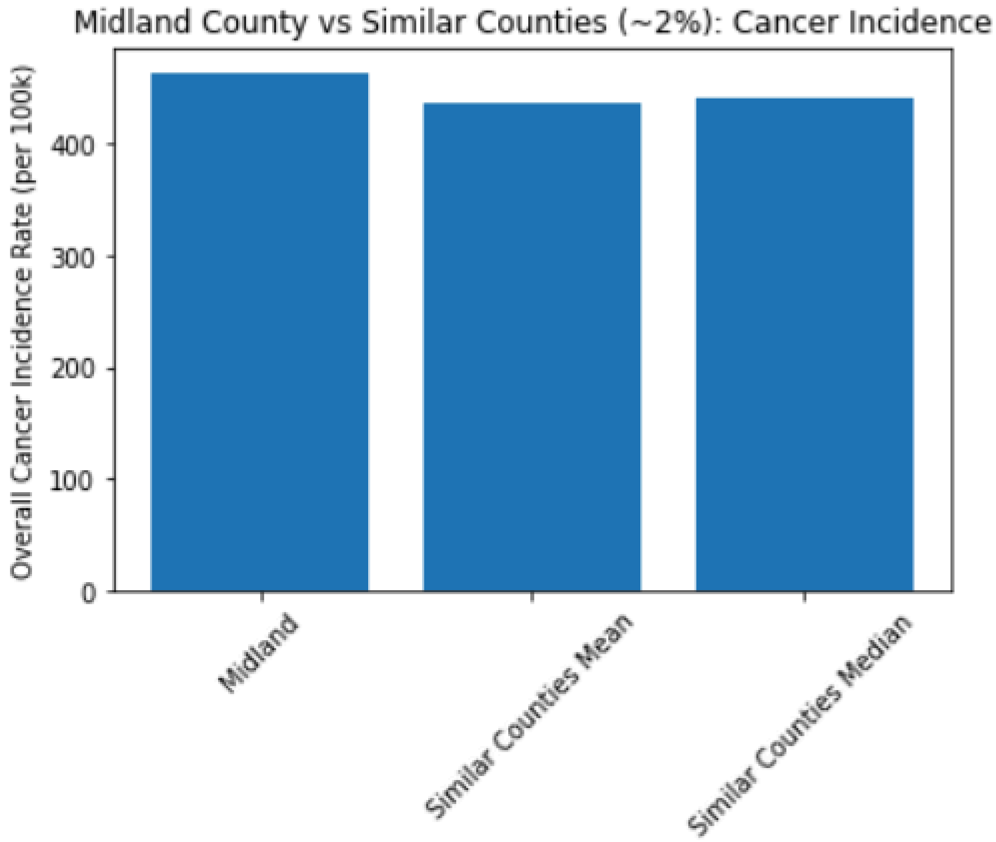
How I Analyzed the Data

To investigate whether Midland county's cancer incidence is elevated compared to counties with similar demographic and socioeconomic makeups, I first identified which counties in Michigan were closest to Midland county. Using the Michigan Cancer Atlas dataset, I selected counties whose percent non-white and poverty rate fell within 2 percent of Midland's percentages. This includes the following counties: Cheboygan, Clinton, Emmet, Grand Traverse, Lapeer, Missaukee, Monroe, and St. Clair. I then calculated the mean cancer incidence rate and the median cancer incidence rate per 100,000 residents in these counties, and compared it to Midland's cancer incidence rate. To test whether Midland's increased rate was statistically significant, I ran a one sample t-test, comparing the similar counties' incidence rates to Midland's.

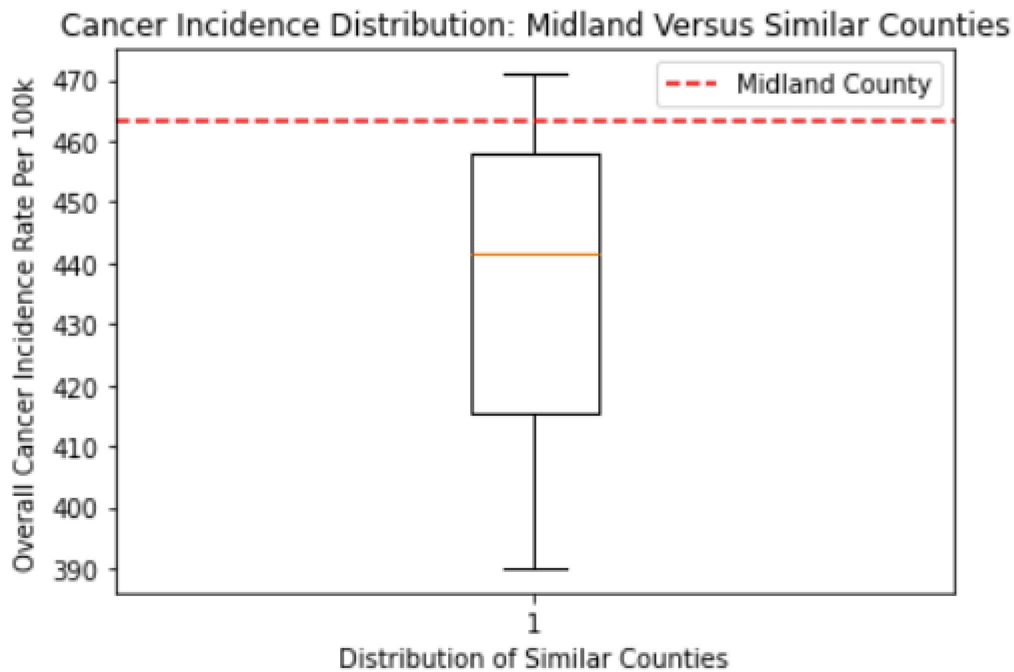
What I found

The results of this analysis showed that the mean cancer rate of the similar counties is approximately 437.0 per 100k, and the median is 441.5 per 100k, which both fall short of Midland's cancer incidence rate of 463.3 per 100k. While that is a noticeable increase, the one sample t-test was a much more meaningful deciding factor on whether these numbers were actually significant. The t-test confirmed that these findings were significant with $t = -2.57$ and a p-value of 0.037, meaning the comparison group's cancer incidence rates are lower than Midland's. To strengthen these results, I widened the similar counties window to 4 percent instead of 2 percent, expanding the comparison group to 30 counties rather than 8. Running the same analysis on this group strengthened the significance of the findings, with $t = -5.89$ and a p-value of less than 0.001. As an additional check, I repeated the same analysis using cancer mortality rates instead of cancer incidence, which actually showed that Midland county's mortality rate is lower than similar counties (within 2 percent) mean and median, though the difference was not as significant, with $t = 1.99$ and $p = 0.087$). This suggests that while Midland residents are diagnosed with cancer at a higher rate, they do not experience a higher mortality rate, which could reflect greater access to healthcare and treatment options. While these results are meaningful, there are still some limitations with this analysis. While poverty rate and percent non-white are major factors in regards to healthcare, this analysis does not take into account access to healthcare, smoking and obesity rates, healthcare quality, and occupational exposure to carcinogen chemicals. Ultimately, this cannot point to carcinogen release as the cause of Midland's elevated cancer incidence rate, although it is alarming.

Visualizations



Midland county's cancer incidence rate of 463.3 per 100k residents exceeds both the mean (437.0 per 100k) and median (441.5 per 100k) cancer incidence rate of eight demographically and economically similar counties.



Midland county's cancer incidence rate falls above the upper quartile of the distribution among demographically and economically similar counties.

Interpretation

These results show strong statistical evidence for what many members of Midland county have long suspected: Midland county's cancer incidence rate is higher compared to demographically and economically similar Michigan counties. The findings are supported by both the 2% comparison group of eight similar counties and the 4% comparison group of thirty similar counties, meaning that the comparison group did not affect the results of this analysis. That being said, this analysis is not enough to link Midland county's elevated cancer incidence rate directly to pollution from the DOW chemical plant, as it only brings in two other variables. The biggest takeaway is that Midland's elevated cancer rate is unlikely to be derived from socioeconomic or demographic factors, which, as established in the previous research question, are some of the biggest drivers in cancer incidence discrepancy across Michigan counties. This strengthens the case for a more in-depth and targeted analysis of Midland county's cancer incidence.

6. Conclusion: Recommendations

Recommendation 1: Promote County-Level Public Health Campaigns targeting Smoking, Obesity, and Nutrition and Healthcare Access For Poorer Populations

Audience: Michigan Department of Health and Human Services and County Health Departments

The analysis conducted in RQ1 ultimately showed that smoking rate, obesity rate, and poverty rates are far stronger predictors of cancer incidence rate than total carcinogen release, with a county's obesity rate having a regression coefficient over 6000 times as large as the total carcinogen release's coefficient. This suggests that a more effective strategy in reducing cancer rates across Michigan counties would be to target community health, lifestyle choices, access to medical care, and access to a healthier diet rather than targeting industrial pollution (although that is still extremely important). Moving forward, Michigan county public health departments and the Michigan Department of Health and Human Services should prioritize campaigns that promote more active lifestyles, refraining from smoking, and access to a balanced diet in poorer regions using existing programs such as SNAP, EBT, and BRIDGE. These campaigns could include social media advertising, flyers in public areas, and even door-to-door public health informants.

Recommendation 2: Launch an Individual-Level Study of Cancer Incidence Among Midland County Residents Living Near the DOW Chemical Plant and DOW Workers With Direct Exposure to Carcinogen Chemicals

Audience: Michigan Department of Health and Human Services, the Environmental Protection Agency, and DOW Chemical

The Findings of RQ2 show that Midland county has a statistically significantly higher cancer incidence rate than demographically and economically similar counties, which held up under both a 2% comparison and a 4% comparison. While these findings are significant, this analysis does not identify exactly why Midland county's cancer incidence is elevated, or if DOW Chemical's carcinogen release has any meaningful contribution. In order to identify whether this relationship is actually causal, the Michigan Department of Health and Human Services and the Environmental Protection Agency should fund a targeted study on populations with proven direct exposure to DOW carcinogens. This includes current and former DOW workers, residents residing near the plant, and residents living in the Tittabawassee River floodplain. The study should include individual medical histories (to determine whether individuals are already at risk for cancer), occupational exposure histories, proximity to contamination sites, and looking at other determinants of cancer seen in RQ1 (obesity, smoking, poverty, etc.). While this study alone would not be entirely conclusive of DOW Chemical's contribution to Midland's elevated cancer incidence, it could justify a targeted cancer screening program or fines imposed on DOW Chemical.